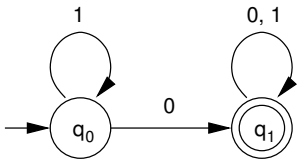
Theory of Computation: Questions and Solutions

Question 1

Consider the following state diagram where the alphabet is $\Sigma = \{0, 1\}$:



Answer the following questions:

1. What is the formal definition of the above machine?

   Solution: The above machine M is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$ where

   - $Q = \{q_0, q_1\}$
   - $\Sigma = \{0, 1\}$
   - δ is given by

	0	1
q_0	q_1	q_0
q_1	q_1	q_1

   - q_0 is the start state
   - $F = \{q_1\}$

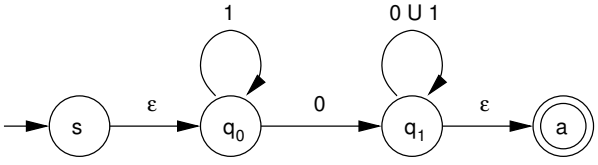
2. What is the language of the above machine?

   Solution: The language of the above machine is the set of all strings that contain a 0. In other words, $\{w \mid w \text{ contains a } 0\}$.

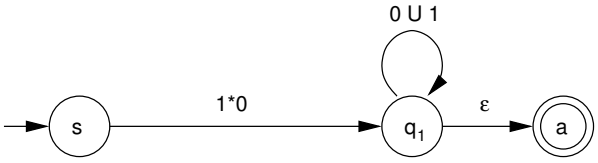
3. What is the regular expression of the above machine?

   Solution: We need to convert the above machine to **generalize nondeterministic finite automaton** and reduce the number of states to 2.

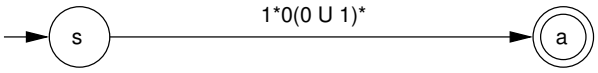
   1. Add new start state s, accept state a, and appropriate ε arrows:

   2. Get rid of the state q_0:

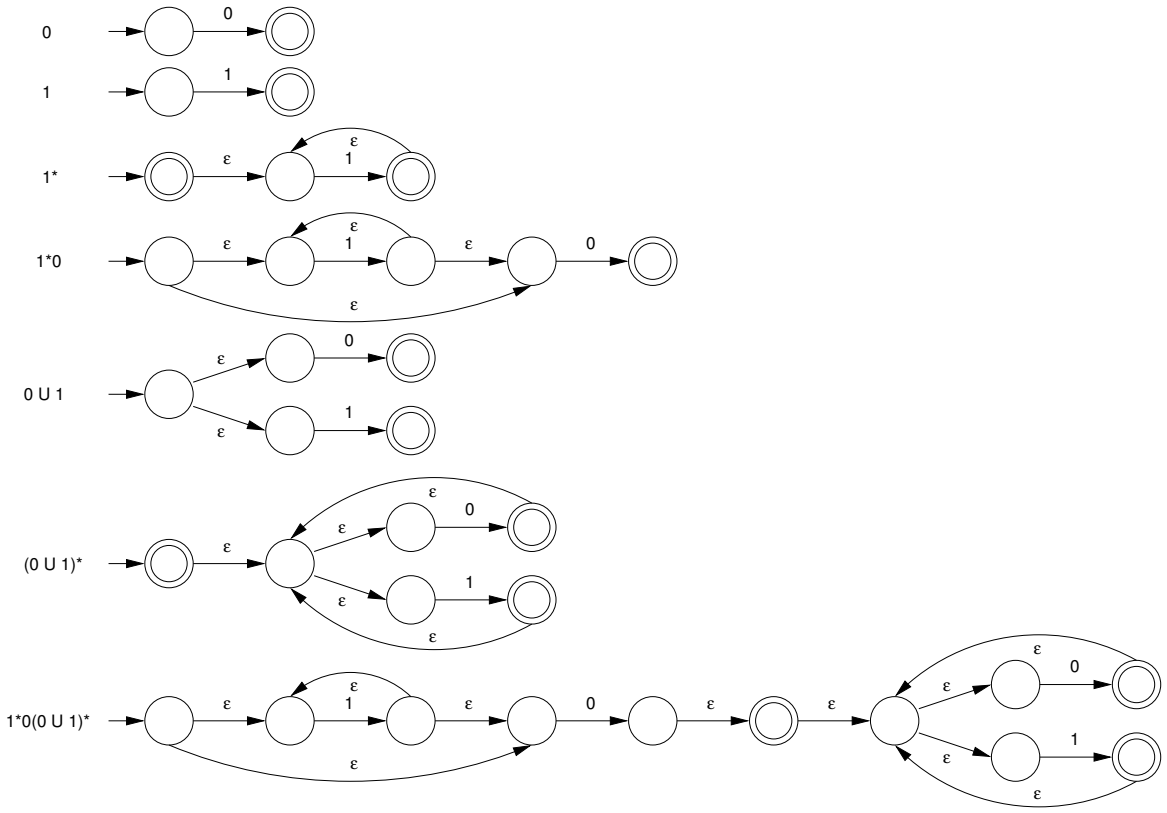
3. Get rid of the state q_1:



Thus, the regular expression is $1^*0(0 \cup 1)^*$.

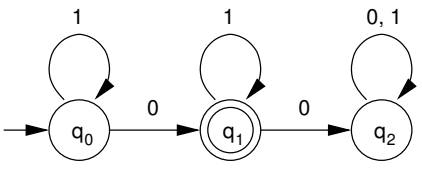
4. From the regular expression in previous question, construct a **nondeterministic finite automaton**

Solution:



Question 2

Consider the following state diagram where the alphabet is $\Sigma = \{0, 1\}$:



Answer the following questions:

1. What is the formal definition of the above machine?

Solution: The above machine M is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$ where

- $Q = \{q_0, q_1, q_2\}$
- $\Sigma = \{0, 1\}$
- δ is given by

	0	1
q_0	q_1	q_0
q_1	q_2	q_1
q_2	q_2	q_2

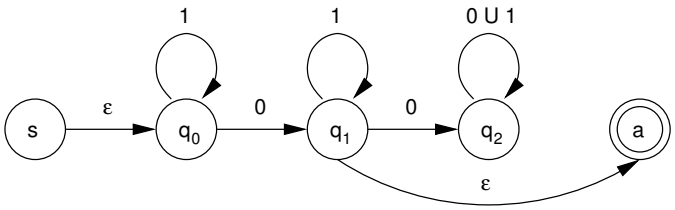
- q_0 is the start state
- $F = \{q_1\}$

2. What is the language of the above machine?

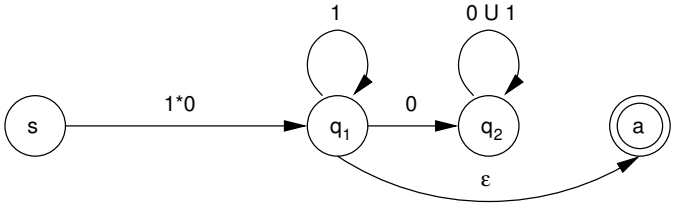
   Solution: The language of the above machine is the set of all strings that contain exactly one 0. In other words, $\{w \mid w \text{ contains exactly one } 0\}$.

3. What is the regular expression of the above machine? **Solution**: We need to convert the above machine to **generalize nondeterministic finite automaton** and reduce the number of states to 2.

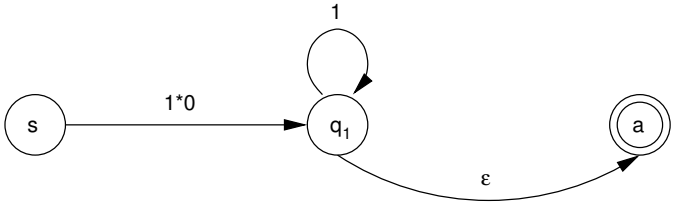
   1. Add new start state s, accept state a, and appropriate ε arrows:



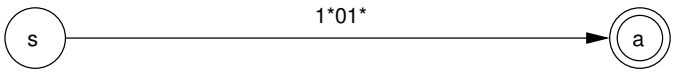
   2. Get rid of the state q_0:



   3. Get rid of the states q_2:



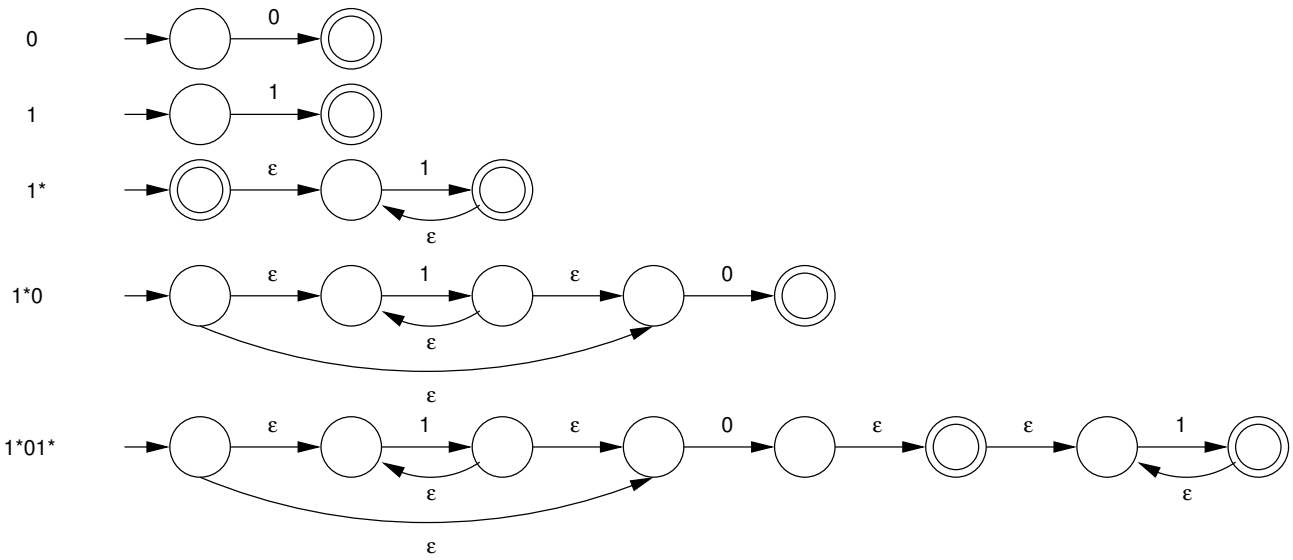
   4. Get rid of the state q_0:



Thus, the regular expression is 1^*01^*.

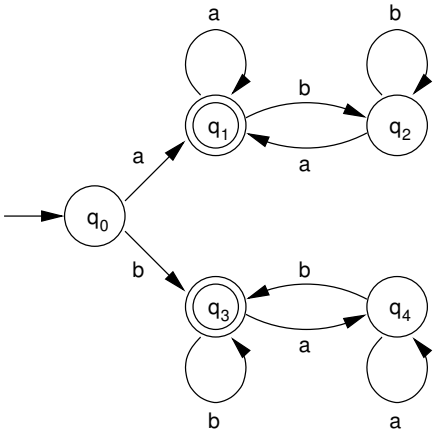
4. From the regular expression in previous question, construct a **nondeterministic finite automaton**

   Solution:



Question 3

Consider the following state diagram where the alphabet is $\Sigma = \{a, b\}$:



Answer the following questions:

1. What is the formal definition of the above machine?

   Solution: The above machine M is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$ where

   - $Q = \{q_0, q_1, q_2, q_3, q_4\}$
   - $\Sigma = \{a, b\}$

- δ is given by

	a	b
q_0	q_1	q_3
q_1	q_1	q_2
q_2	q_1	q_2
q_3	q_4	q_3
q_4	q_4	q_3

- q_0 is the start state
- $F = \{q_1, q_3\}$

2. What is the language of the above machine?

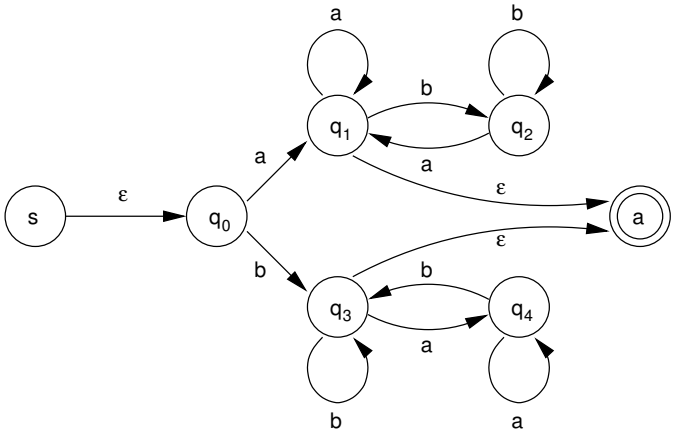
   Solution: The language of the above machine is the set of all strings that start and end with the same symbol. In other words,

   $$\{w \mid w \text{ starts and ends with the same symbol}\}$$

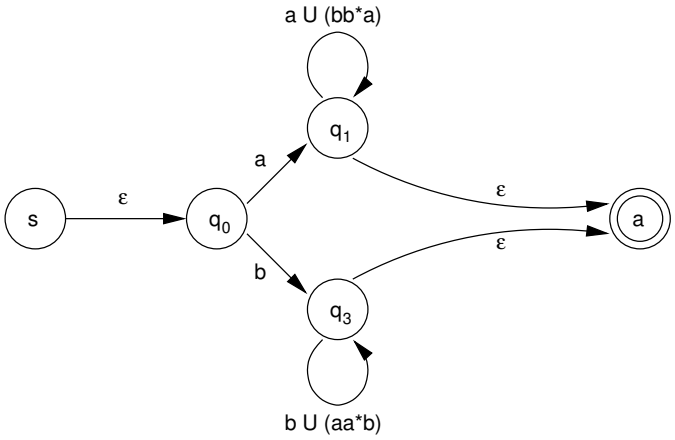
3. What is the regular expression of the above machine?

   Solution: We need to convert the above machine to **generalize nondeterministic finite automaton** and reduce the number of states to 2.

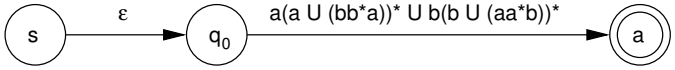
   1. Add new start state s, accept state a, and appropriate ε arrows:



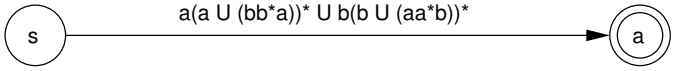
   2. Get rid of the states q_2 and q_4:



3. Get rid of the states q_1 and q_3:



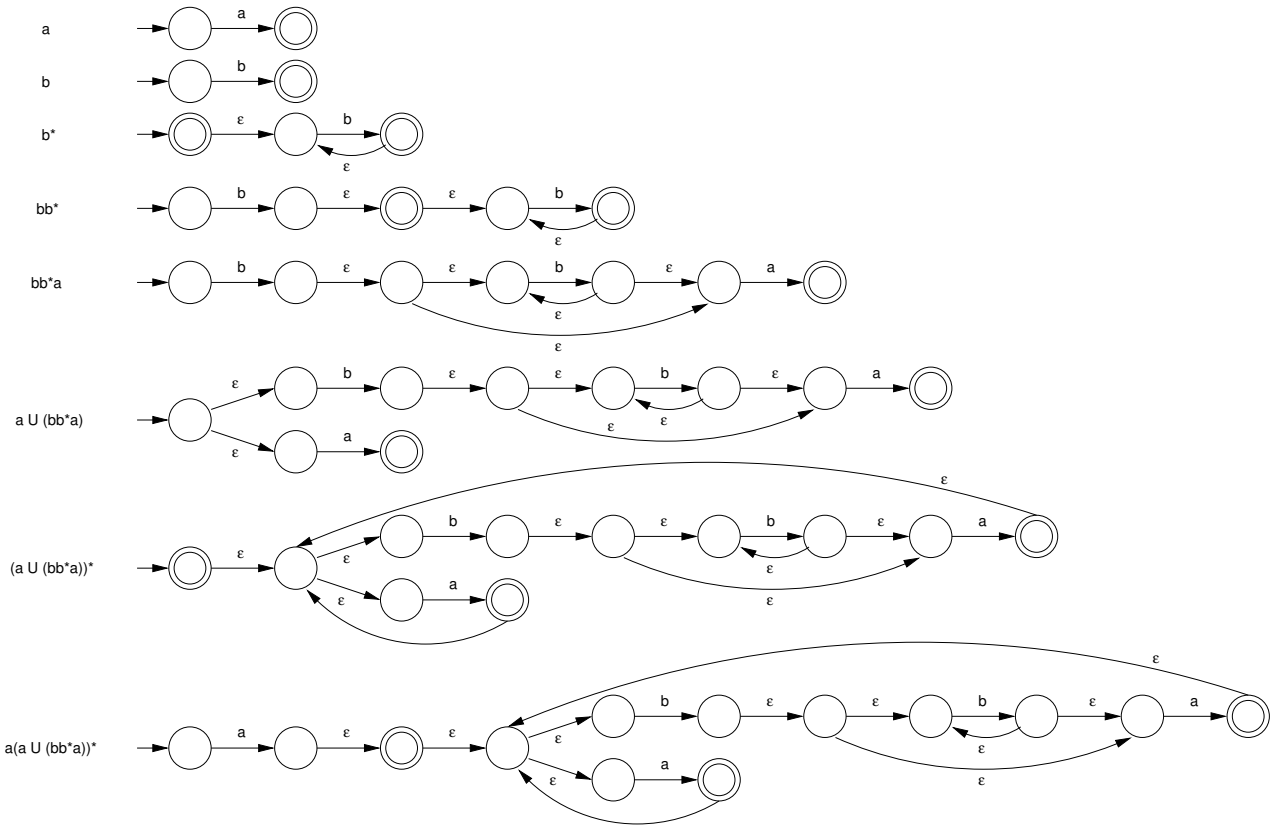
4. Get rid of the state q_0:



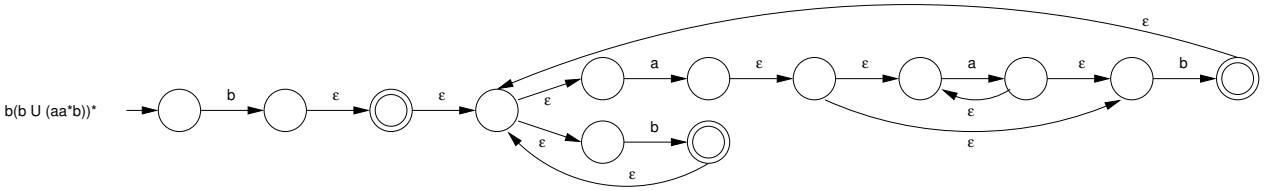
Thus, the regular expression is $a(a \cup (bb^*a))^* \cup b(b \cup (aa^*b))^*$.

4. From the regular expression in previous question, construct a **nondeterministic finite automaton**

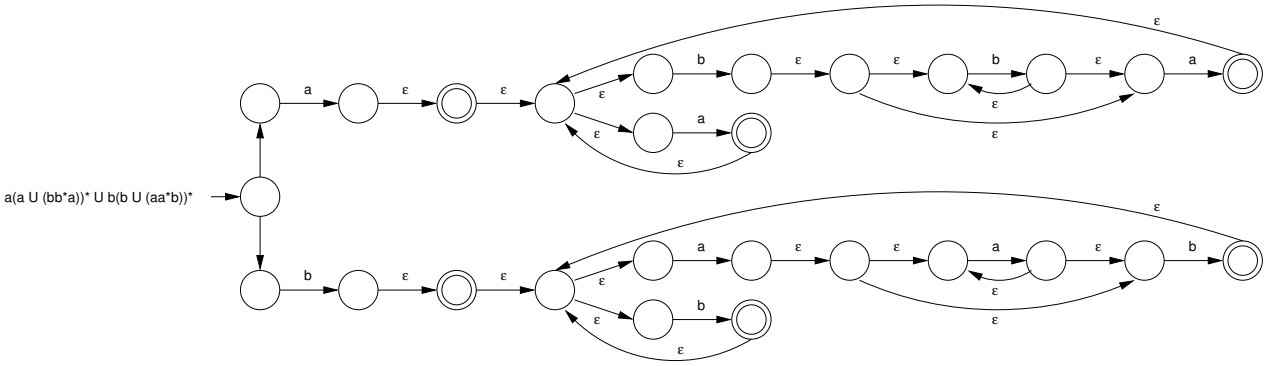
Solution: First, we construct the NFA for $a(a \cup (bb^*a))^*$:



Next, we construct the NFA for $b(b \cup (aa^*b))*$ which is similar to $a(a \cup (bb^*a))^*$:

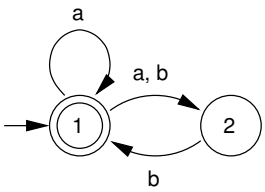


Then we construct $a(a \cup (bb^*a))^* \cup b(b \cup (aa^*b))^*$



Question 4

Consider the following state diagram where the alphabet is $\Sigma = \{a, b\}$:



Note that the above machine is a **nondeterminisitc finite automaton**. Answer the following questions:

1. What is the formal definition of the above machine?

   Solution: The above machine M is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$ where

   - $Q = \{1, 2\}$
   - $\Sigma = \{a, b\}$
   - δ is given by

	a	b
1	$\{1, 2\}$	$\{2\}$
2	$\emptyset$	$\{1\}$

   - 1 is the start state
   - $F = \{1\}$

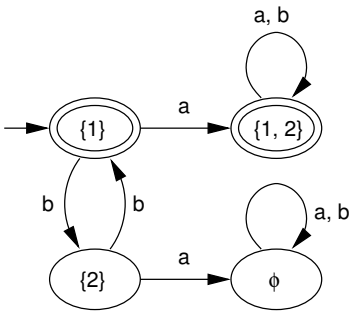
2. Convert the above machine to equivalent deterministic finite automata

   Solution: The equivalent machine M' is a 5-tuple $(Q', \Sigma, \delta', q_0', F')$ where

   - $Q' = \mathcal{P}(Q) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$
   - $\Sigma = \{a, b\}$
   - δ' is given by

	a	b
$\emptyset$	$\emptyset$	$\emptyset$
$\{1\}$	$\{1, 2\}$	$\{2\}$
$\{2\}$	$\emptyset$	$\{1\}$
$\{1, 2\}$	$\{1, 2\}$	$\{1, 2\}$

   - $\{1\}$ is the start state
   - $F' = \{\{1\}, \{1, 2\}\}$



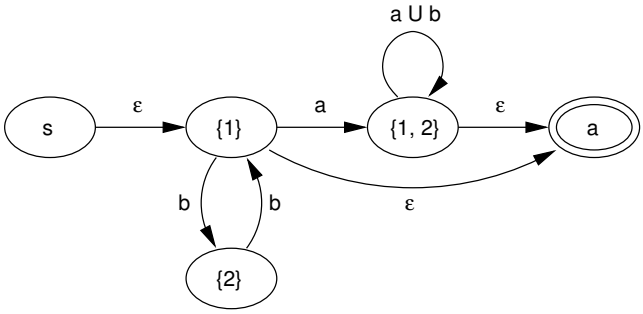
3. What is the language of this machine?

   Solution: The language of this machine is an empty string, set of all strings that contain a symbol a, or set of all strings that starts with nothing but even number of b.

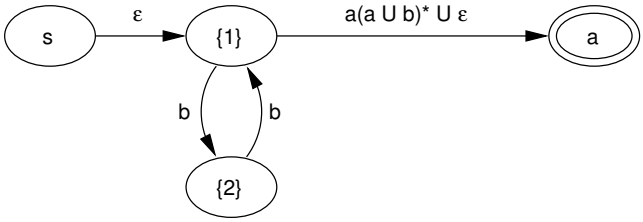
4. What is the regular expression of the above machine?

   Solution: We need to convert the above machine to **generalize nondeterministic finite automaton** and reduce the number of states to 2.

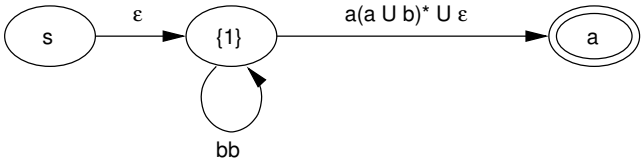
   1. Add new start state s, accept state a, and appropriate ε arrows:



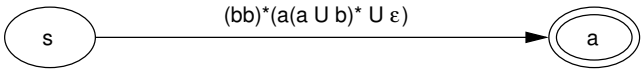
   2. Get rid of the states $\{1, 2\}$:



   3. Get rid of the states $\{2\}$:



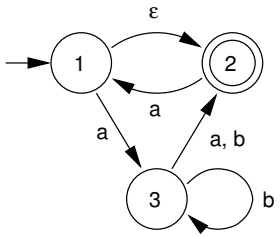
   4. Get rid of the state $\{1\}$:



Thus, the regular expression is $(bb)^*(a(a \cup b)^* \cup \varepsilon)$.

Question 5

Consider the following state diagram where the alphabet is $\Sigma = \{a, b\}$:



Note that the above machine is a **nondeterminisitc finite automaton**. Answer the following questions:

1. What is the formal definition of the above machine?

   Solution: The above machine M is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$ where

   - $Q = \{1, 2, 3\}$
   - $\Sigma = \{a, b\}$
   - δ is given by

	a	b	ε
1	$\{3\}$	$\emptyset$	$\{2\}$
2	$\{1\}$	$\emptyset$	$\emptyset$
3	$\{2\}$	$\{2, 3\}$	$\emptyset$

   - 1 is the start state
   - $F = \{2\}$

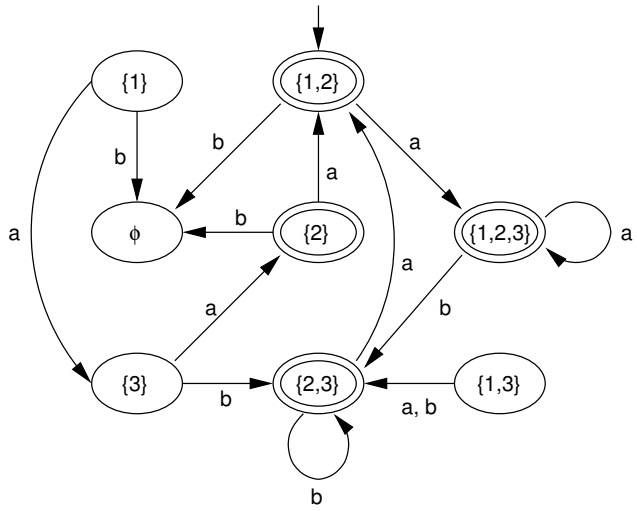
2. Convert the above machine to equivalent deterministic finite automata

   Solution: The equivalent machine M' is a 5-tuple $(Q', \Sigma, \delta', q_0', F')$ where

   - $Q' = \mathcal{P}(Q) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$
   - $\Sigma = \{a, b\}$
   - δ' is given by

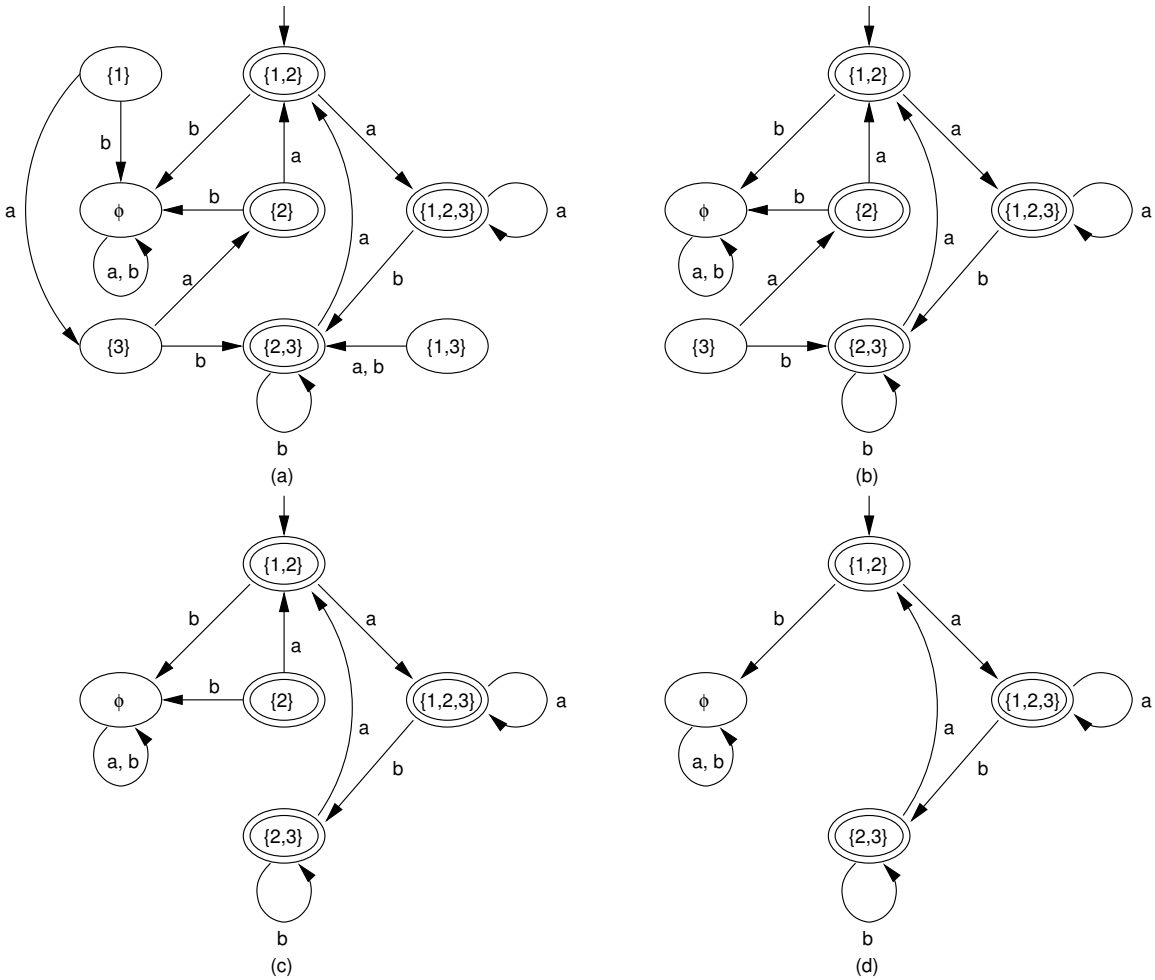
	a	b
$\emptyset$	$\emptyset$	$\emptyset$
$\{1\}$	$\{3\}$	$\emptyset$
$\{2\}$	$\{1,2\}$	$\emptyset$
$\{3\}$	$\{2\}$	$\{2,3\}$
$\{1,2\}$	$\{1,2,3\}$	$\emptyset$
$\{1,3\}$	$\{2,3\}$	$\{2,3\}$
$\{2,3\}$	$\{1,2\}$	$\{2,3\}$
$\{1,2,3\}$	$\{1,2,3\}$	$\{2,\ 3\}$

- $\{1,2\}$ is the start state
- $F' = \{\{2\},\{1,2\},\{2,3\},\{1,2,3\}\}$



3. What is the language of this machine?

   Solution: It is quite difficult to figure out what is the language of the machine above. However, there are some states that cannot be reach from start state. These states can be eliminated. See Figure.

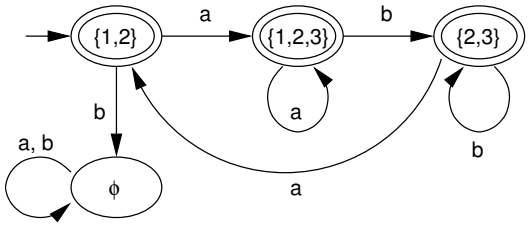


- In Figure (a), states $\{1\}$ and $\{1,3\}$ cannot be reached by any other state. We can remove the state $\{1\}$. Result in Figure (b).
- In Figure (b), state $\{3\}$ cannot be reached by any other state. We can remove the state $\{3\}$. Result in Figure (c).
- In Figure (c), state $\{2\}$ cannot be reached by any other state. We can remove the state $\{2\}$. Result in Figure (d).

From Figure (d), the language of this machine is a set of all strings that do not start with a 'b' and do not contain bab as a substring.

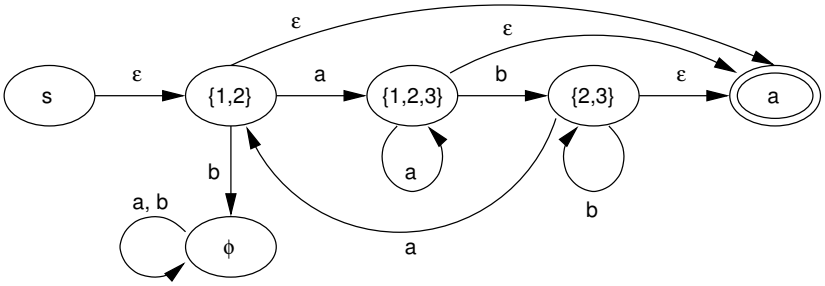
4. What is the regular expression of the above machine?

   Solution: Let's rearrange the state diagram a little bit before we start. The state diagram below is the same as in Figure (d):

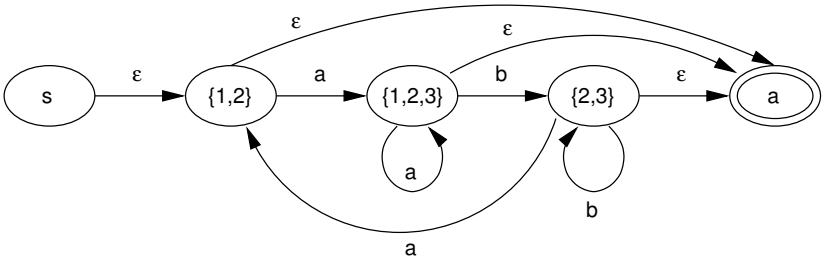


We need to convert the above machine to **generalize nondeterministic finite automaton** and reduce the number of states to 2.

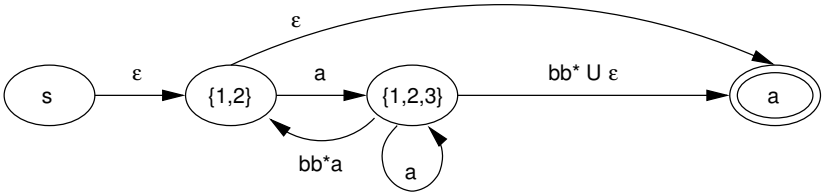
1. Add new start state s, accept state a, and appropriate ε arrows:



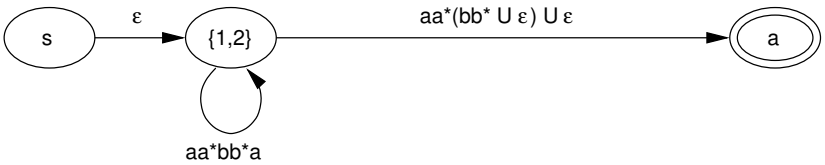
2. Get rid of the states $\{\emptyset\}$:



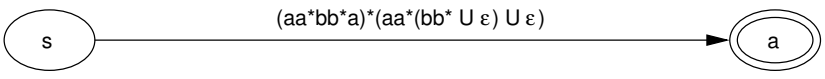
3. Get rid of the states $\{2, 3\}$:



4. Get rid of the state $\{1, 2, 3\}$:



5. Get rid of the state $\{1, 2\}$:

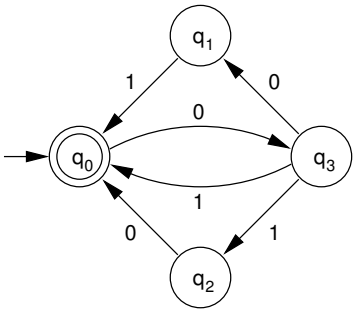


Thus, the regular expression is $(aa^*bb^*a)^*(aa^*(bb^* \cup \varepsilon) \cup \varepsilon)$.

Question 6

1. Give an NFA recognizing the language $(01 \cup 001 \cup 010)^*$

   Solution:



   The formal definition of the above machine is 5-tuple $(Q, \Sigma, \delta, q_0, F)$ where

   - $Q = \{q_0, q_1, q_2, q_3\}$
   - $\Sigma = \{0, 1\}$
   - δ is given by

	0	1
q_0	$\{q_3\}$	$\emptyset$
q_1	$\emptyset$	q_0
q_2	$\{q_0\}$	$\emptyset$
q_3	$\{q_1\}$	$\{q_0, q_2\}$

   - q_0 is the start state
   - $F = \{q_0, q_5\}$

2. Convert this NFA to an equivalent DFA. Give only the portion of the DFA that is reachable from the start state.

   Solution: The equivalent DFA M' of 5-tuple $(Q', \Sigma, \delta', q_0', F')$ can be constructed by

   - Set of states:

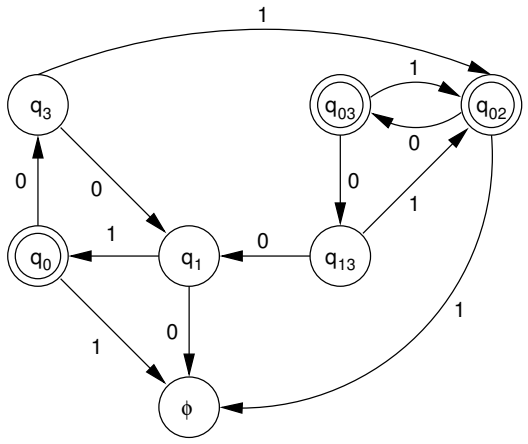
$$\begin{aligned} Q' = \mathcal{P}(Q) = \{&\emptyset, \{q_0\}, \{q_1\}, \{q_2\}, \{q_3\}, \{q_0, q_1\}, \{q_0, q_2\}, \{q_0, q_3\}, \\ &\{q_1, q_2\}, \{q_1, q_3\}, \{q_2, q_3\}, \{q_0, q_1, q_2\}, \{q_0, q_1, q_3\} \\ &\{q_0, q_2, q_3\}, \{q_1, q_2, q_3\}, \{q_0, q_1, q_2, q_3\}\} \end{aligned}$$

   - $\Sigma = \{0, 1\}$
   - δ' is given by

	0	1
$\emptyset$	$\emptyset$	$\emptyset$
$\{q_0\}$	$\{q_3\}$	$\emptyset$
$\{q_1\}$	$\emptyset$	$\{q_0\}$
$\{q_2\}$	$\{q_0\}$	$\emptyset$
$\{q_3\}$	$\{q_1\}$	$\{q_0, q_2\}$
$\{q_0, q_1\}$	$\{q_3\}$	$\{q_0\}$
$\{q_0, q_2\}$	$\{q_0, q_3\}$	$\emptyset$
$\{q_0, q_3\}$	$\{q_1, q_3\}$	$\{q_0, q_2\}$
$\{q_1, q_2\}$	$\{q_0\}$	$\{q_0\}$
$\{q_1, q_3\}$	$\{q_1\}$	$\{q_0, q_2\}$
$\{q_2, q_3\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
$\{q_0, q_1, q_2\}$	$\{q_0, q_3\}$	$\{q_0\}$
$\{q_0, q_1, q_3\}$	$\{q_1, q_3\}$	$\{q_0, q_2\}$
$\{q_0, q_2, q_3\}$	$\{q_0, q_1, q_3\}$	$\{q_0, q_2\}$
$\{q_1, q_2, q_3\}$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
$\{q_0, q_1, q_2, q_3\}$	$\{q_0, q_1, q_3\}$	$\{q_0, q_2\}$

- $\{q_0\}$ is the start state
- $F' = \{\{q_0\}, \{q_0, q_1\}, \{q_0, q_2\}, \{q_0, q_3\}, \{q_0, q_1, q_2\}, \{q_0, q_1, q_3\}, \{q_0, q_2, q_3\}, \{q_0, q_1, q_2, q_3\}\}$

Note that from the above transition function, the only states that can be reached from the start state are $\emptyset$, $\{q_0\}$, $\{q_1\}$, $\{q_3\}$, $\{q_0, q_2\}$, $\{q_0, q_3\}$, and $\{q_1, q_3\}$.



Pumping Lemma

Pumping lemma: If A is a regular language, then there is a number p (the pumping length) where if s is any string in A of length at least p, then s may be divided into three pieces, $s = xyz$, satisfying the following conditions:

1. for each $i \geq 0$, $xy^iz \in A$,
2. $|y| > 0$, and

3. $|xy| \leq p$.

Example 1

Show that $B = \{0^n 1^n \mid n \geq 0\}$ is not regular.

Solution: Assume that B is regular. Since B is regular, the Pumping lemma says that for any string $s \in B$ of length at least p can be divided into three pieces, $s = xyz$, satisfying the following three conditions:

1. for each $i \geq 0$, $xy^i z \in A$,
2. $|y| > 0$, and
3. $|xy| \leq p$.

Let $s = 0^p 1^p$. Since $s \in A$ and s has length at least p, the Pumping lemma says that s can be divided into $s = xyz$, satisfying the above three conditions. To satisfy the second and the third conditions, $x = 0^j$ and $y = 0^k$ for some j and k where $j \geq 0$, $k > 0$, and $j + k \leq p$. Since $xy = 0^j 0^k$ where $j + k \leq p$, z must be $0^n 1^p$ where $n = p - (j + k)$. Let's check whether xyz is equivalent to s:

$$\begin{aligned} xyz &= 0^j 0^k 0^n 1^p \\ &= 0^j 0^k 0^{p-(j+k)} 1^p \\ &= 0^{j+k+p-(j+k)} 1^p \\ &= 0^p 1^p \end{aligned}$$

Let's check whether $xy^0 z \in A$:

$$\begin{aligned} xy^0 z &= 0^j (0^k)^0 0^n q^p \\ &= 0^j (0^k)^0 0^{p-(j+k)} 1^p \\ &= 0^j 0^{0 \times k} 0^{p-(j+k)} 1^p \\ &= 0^j 0^0 0^{p-(j+k)} 1^p \\ &= 0^j \varepsilon 0^{p-(j+k)} 1^p \\ &= 0^j 0^{p-(j+k)} 1^p \\ &= 0^{j+p-(j+k)} 1^p \\ &= 0^{p-k} 1^p \end{aligned}$$

Since $k > 0$, $p - k \neq p$. In other words, the number of 0s in the string $xy^0 z$ is less than the number of 1s. Thus, $xy^0 z \notin A$ which contradicts with the first condition.